

1st REVIEW

Online Auction and Bidding Platform

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Under the Guidance
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GROUP NUMBER:17

Name of Candidate	Role
U.Moksha	Objectives + whatever slides explain how the system works/features
B.Jamuna	Introduction, Problem Statement
Dhana sri	Technical implementation, database, modules, screenshots/demo, results, conclusion

ABSTRACT

- The **Online Auction and Bidding Platform** is a web-based system designed to provide a secure, transparent, and efficient environment for buying and selling products through online auctions. The platform allows sellers to create and manage auction listings by providing details such as product description, starting price, and auction duration. Buyers can view available products and place bids in real time.
- The system uses **Socket.io** for instant bid updates, ensuring that users receive the latest bidding information without delays. An **anti-sniping mechanism** is implemented to reduce unfair last-second bidding and provide a fair auction experience. **MySQL** is used to securely manage users, auctions, bids, payments, and transaction data. The platform also supports secure online payments through **Stripe**, along with separate dashboards for buyers, sellers, and administrators.
- The proposed system aims to simplify the complete auction process, from product listing and live bidding to winner selection, payment, and final settlement. Overall, the platform provides a reliable, user-friendly, and transparent solution for conducting online auctions.

INTRODUCTION

- The **Online Auction & Bidding Platform** is a web-based system designed to provide a secure and efficient environment for buying and selling products through online auctions. Sellers can list products with details such as starting price, description, and auction duration, while buyers can view listings and place bids in real time.
- The system uses **MySQL** to manage important data such as users, auctions, bids, payments, and transactions. **Socket.io** enables real-time bid updates, while an **anti-sniping mechanism** extends the auction when bids are placed near the closing time. The platform also supports secure payment processing, seller and buyer dashboards, auction verification, and dispute management.
- The main goal of the project is to provide a **transparent, reliable, and user-friendly auction platform** that manages the complete process from product listing and live bidding to winner selection, payment, and settlement.

EXISTING SYSTEM

- Traditional auction systems often depend on **manual bidding or basic online bidding mechanisms**.
- Bid updates may not be **instantaneous**, causing delays and confusion among users.
- **Last-second bidding** can give an unfair advantage to some participants.
- Auction listings, bids, and transactions may be managed separately, making **data management difficult**.
- Buyers and sellers may have limited dashboards for tracking their activities.
- Payment and settlement processes may require **manual verification**.
- There may be limited support for **dispute management and transaction tracking**.
- These limitations can make the auction process **less efficient, transparent, and user-friendly**.

PROBLEM

- Lack of **real-time bid updates** can cause delays during auctions.
- **Last-second bidding** may result in an unfair advantage.
- Managing large numbers of **users, auctions, bids, and transactions** can be difficult.
- Manual payment and settlement processes can lead to **errors and delays**.
- Buyers and sellers may not have proper **dashboards to track their activities**.
- Lack of proper **auction verification and dispute management** can reduce trust.
- A centralized and secure database is required to maintain **accurate and consistent auction data**.

SOLUTION(Contribution)

- Use **MySQL** to securely store users, products, auctions, bids, payments, and transaction records.
- Use **ER modeling, normalization, constraints, indexes, views, and SQL queries** to maintain an efficient database.
- Use **Node.js and Express.js** to develop REST APIs for auction and user management.
- Use **JWT authentication and RBAC** to provide secure access for buyers, sellers, and administrators.
- Use **Socket.io** for real-time bid updates without refreshing the page.
- Use **Redis** to manage frequently accessed auction and bidding data efficiently.

Objectives

- To develop an online platform for product auctions.
- To provide real-time bidding functionality.
- To implement an anti-sniping mechanism.
- To securely store users, auctions, bids, and transactions.
- To provide secure online payment and settlement.
- To provide dashboards for buyers, sellers, and administrators.
- To provide a dispute management system

Modules

- **Buyer**
 - Browse auctions
 - Place bids
 - View bid history
 - Watchlist
 - Make payments
 - View won auctions
- **Seller**
 - Create auctions
 - Manage products
 - Monitor bids
 - View sold items
 - Track earnings
- **Admin**
 - Verify listings
 - Manage users
 - Monitor auctions
 - Manage disputes
 - Monitor transactions

SOFTWARE REQUIREMENTS

- **Operating System:** Windows / Linux / macOS
- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MySQL
- **ORM:** Sequelize
- **Real-Time Communication:** Socket.io
- **Caching:** Redis
- **Authentication:** JWT
- **API Testing:** Postman
- **Code Editor:** Visual Studio Code
- **Version Control:** Git & GitHub
- **Containerization:** Docker, Docker Compose
- **Browser:** Google Chrome / Microsoft Edge
- **Testing:** Jest / Supertest

CONCLUSION AND FUTURE SCOPE

The **Online Auction & Bidding Platform** provides a secure and efficient solution for conducting online auctions. The system uses **MySQL** to manage users, auctions, bids, and transactions, while **Node.js** and **Express.js** handle the backend services. **Socket.io** enables real-time bidding, and **Redis** improves the performance of frequently accessed data. JWT authentication and role-based access control provide secure access for buyers, sellers, and administrators. Overall, the system provides a transparent and fair auction process from product listing to winner selection and settlement.

Future Scope

- **Microservices Architecture** – Separate auction, user, payment, and notification services for better scalability.
- **Advanced Analytics** – Provide bidding, sales, and user activity reports.
- **Mobile Application** – Extend the platform to Android and iOS.
- **AI-Based Fraud Detection** – Identify suspicious bidding patterns and activities.
- **Automated Notifications** – Add email/SMS notifications for bids, auction endings, and payments.
- **Kubernetes Deployment** – Support scalable cloud deployment.
- **Advanced Monitoring** – Use tools such as Prometheus and Grafana for system monitoring.
- **Improved Testing & CI/CD** – Automate testing, security checks, and deployment using GitHub Actions.

THANK YOU